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Kestrel Box: Solar Panel Sizing

Load Profile

The Raspberry Pi Zero 2 W has an input power of 5V DC 2.5 A, 12.5 W maximum [1]. A resource states the idle current is around 0.1 A, so our idle power is 0.5 W [2]. Assuming constant usage, the maximum and idle yearly energy usage are 109.50 kWh/yr and 4.38 kWh/yr respectively.

Predicting Panel Size

The equation

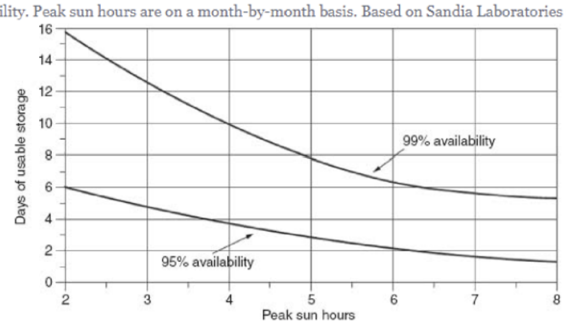
$$Energy\ \left(\frac{kWh}{yr}\right) = P_{DC-STC} * Derate\ Factor * \left(\frac{h}{d}\right) peak\ sun * 365\left(\frac{d}{yr}\right)$$

relates total energy usage to the size of the panel that is required. The energy is already calculated. P is the power rating of the panel. Derate Factor is an efficiency parameter, which 0.85 will be assumed. Peak sun can be obtained from data or calculated. From the NLR PVWatts calculator [3], the lowest peak sun from April - July is around 6 h/d. The calculated rated panel is 58.8 W using the maximum power usage and 2.35 W using the idle power usage. The range of the panel size is quite big, 2.35 W – 58.8 W, but this is using the extreme values of the load. A panel size of around 10 W should be reasonable.

Final Remarks

There is always the chance of having low peak sun (ie. from shading, clouds), and batteries are important to cushion for this. I included an image to give an idea of how a bigger battery decreases the chances of the system dying.

FIGURE 6.31 Days of battery storage needed for a stand-alone system with 95% and 99% system availability. Peak sun hours are on a month-by-month basis. Based on Sandia Laboratories (1995).



It should be made sure that if the system does run out of battery, that it will still operate like normal after turning back on (I don't much of how this works). Lastly, it could be helpful to look at ways of controlling power usage (ie. sleep mode) to work around the unreliability of PV systems.

References

[1] https://pip-assets.raspberrypi.com/categories/584-raspberry-pi-zero-2-w/documents/RP-008359-DS-1-raspberry-pi-zero-2-w-product-brief.pdf?disposition=inline

[2] https://andrejacobs.org/b24/electronics/raspberry-pi-zero-2-w-temperature-and-power-consumption/

[3] https://pvwatts.nlr.gov/pvwatts.php